

SHiP Timing Bar:
Corrected Transport, Measured Light Collection
What is established after the optical-physics audit

René Ríos
Universidad de La Serena (ULS)

September 2026 — v7

The optical-transport baseline has changed

The world-escape guard destroyed reflected photons. The air-wrap border selected a dielectric factory whose measured reflection fraction did not match its nominal input.

Both physics fixes are in main 3dc2c37: guard 218241a, reflector f4d90a6.

- This talk presents measured boundary outcomes, terminal accounting and light-collection comparisons.
- The timing-resolution results in v6 have no corrected replacements yet.
- No historical timing value is promoted to a new result here.

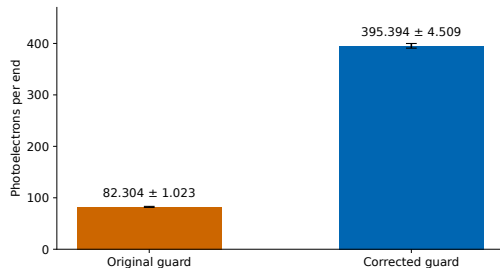
Verified main merge and source genealogy: docs/DEFECTS_AND_CORRECTIONS.md. Measurement configurations and sidecar identifiers accompany each result.

The old rule killed optical tracks by the post-step name WorldLV, without consulting the boundary status.

Observed at BarPV → WorldPV

4930224 encounters;
3750479 were TIR: 76.1%.

A reflected step can still name WorldPV as its post-step volume. The corrected guard checks the boundary status.



$N = 500$ in *both* arms; error bars are event-level SEM.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. EndTop, $N_{\text{TOP}} = 70$, $x = 0$ mm, $N = 500$ /arm, one worker. R1/R2: 0336ba9/218241a; sidecars sources/guard, sources/bar_world.

dielectric_dielectric + polished at the
air-wrap border returned only

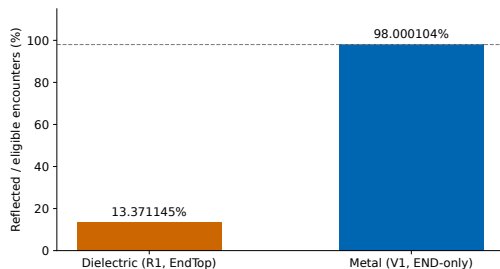
13.371145%

of eligible forward encounters. The metal control
returned

98.000104%.

Interpretation in the tested Geant4 11.4.0

build: the dielectric input admits survival before
Fresnel; it does not directly set the reflected
fraction. Source mechanism is documented as a
hypothesis in backup.



Different geometries are labeled. This is a
boundary-response comparison, not a matched
yield ratio.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. $x = 0$ mm. R1: EndTop, $N_{TOP} = 70$, $N = 500$, one worker, 0336ba9. V1: END-only, $N_{TOP} = 0$, $N = 2000$, 12 workers/eventModulo 1, 2e3872e. Sidecars: sources/directional, sources/metal_reflection.

Scintillation tracks started	40039535
Terminal scintillation tracks	40039535
Residual	0%

Disjoint accounting

Each track contributes once at termination.
Duplicate terminals, unclassified origins and the start-terminal residual are zero in this run.

Ledger closure validates accounting; it does not by itself validate the optical model.

Terminal category	Tracks
Bulk absorption in bar	9099837
Bulk absorption in wrap	197707
Explicit world guard	8553596
Other terminal transport/boundary	22188395

Other transport/boundary is the remaining terminal group. It is not equated to the hit count or to detection alone.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. EndTop, $N_{\text{TOP}} = 70$, $x = 0$ mm, $N = 2000$, one worker. D1: 2ddf205; sidecars sources/terminal_scint, sources/terminal_audit.

Configuration	Metal / dielectric END yield	Comparison
END-only	1.09896 ± 0.00852	D0/D3
EndTop	1.01155 ± 0.00830	B0/A0
Ratio of those gains	1.08641 ± 0.01227	factorial contrast

The guard is already corrected in these comparisons

The reflector is the changed optical choice within each readout geometry. A single multiplicative “reflector correction” does not describe both geometries.

Ratio uncertainties retain the historical zero-covariance approximation; using the same seeds does not supply an estimated covariance.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. $x = 0$ mm, $N = 2000$ /arm. END-only: D0/D3, $N_{\text{TOP}} = 0$, 2ddf205/391fa4e; EndTop: B0/A0, $N_{\text{TOP}} = 70$, f4d90a6/a7a9090. Worker settings in backup; sidecars sources/factorial.

Many more encounters accompany a modest END-yield gain

END-only factory	Air $\rightarrow$ wrap encounters / started optical photon
Dielectric, D3	0.48896774
Metal, V1	10.66476630

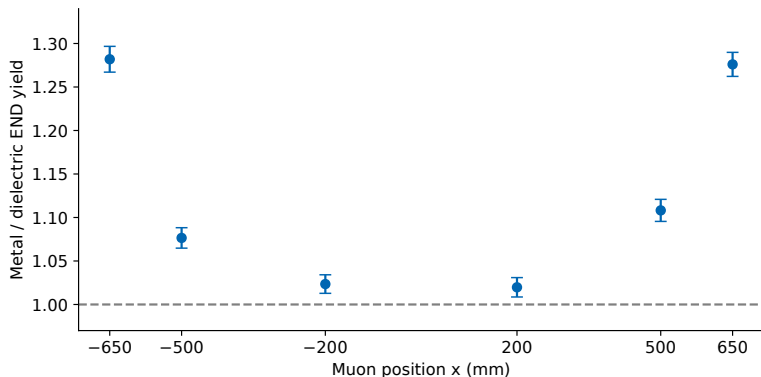
Measured: about $21.8\times$ more forward encounters

Encounters include outcomes other than reflection. These are unconditional all-optical means, not means per scintillation photon and not counts of successful bounces.

Limited interpretation: specular reflection preserves the incidence angle for equal or parallel face normals. Recycling need not restore TIR acceptance at those faces. Orthogonal faces differ; no angular-history study was performed.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. END-only, $N_{\text{TOP}} = 0$, $x = 0$ mm, $N = 2000/\text{arm}$, 12 workers/eventModulo 1. D3/V1: 391fa4e/2e3872e. Sidecars: sources/multiplicity_dielectric, sources/multiplicity_metal.

EndTop: the reflector gain grows toward the ends



The comparison uses the same readout layout and a corrected guard in both arms. Error bars are the recorded approximate ratio SEMs.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. EndTop, $N_{\text{TOP}} = 70$; $x \in \{\pm 200, \pm 500, \pm 650\}$ mm; $N = 1000/\text{arm/position}$, one worker. Dielectric/metal commits: a7a9090/f4d90a6. Sidecars: sources/positions, figs/position_gain. Per-cell commands: sources/provenance.json.

Matched design comparison:

D1: EndTop with an uncovered upper face around TOP sensors.

D2: add the upper wrap panel while retaining TOP.

Measured END yield ratio

$$D2/D1 = 1.03244 \pm 0.00869$$

This is an experimental design measurement, not a production correction or an established optimum.

Baseline TOP hit/generation ratio:

$$\frac{9510815}{40039535} = 23.75\%$$

Numerator: all detected TOP hits.

Denominator: generated scintillation photons.

This mixed-population ratio is not a measured fraction of unique scintillation photons intercepted by TOP. It does not establish an END depletion fraction or a timing penalty.

EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. EndTop, $N_{\text{TOP}} = 70$, $x = 0$ mm, $N = 2000/\text{arm}$, one worker. D1/D2: 2ddf205/8b64721. Sidecars: sources/design; TOP/generation counts in the D1 summary linked by sources/provenance.json.

What the historical claims audit establishes

Audit status	Rows	Meaning in the historical inventory
valid	180	Source value or computation reproduced; not corrected-physics validation.
unverified	778	Provenance or interpretation not established.
superseded	7	Replaced as a source-of-truth entry in the historical record.
refuted	2	Contradicted in the stated context.
Total	967	Audited source occurrences, not independent measurements.

Scope of this talk

The inventory is preserved. No historical timing result is presented as a measurement with the corrected physics.

Historical inventory: CLAIMS_AUDIT_20260910.md, before deck edits, source main 8349041. Local CSV/ROOT/metadata: sources/claims_audit. Configurations vary by row; no single event sample or seed pair applies to audit counts.

Historical numbers: source evidence is not a replacement result

Number quoted for audit	Provenance	Status
{1310.8, 941.0, 701.3} pe/end	Literal <code>sim_npe</code> dictionary in <code>napkin.py</code> :71. Material order follows the archived source.	No registered run provenance at that assignment; not a new simulation measurement.
52.1 ps	<code>sigma_sim_ej204</code> in the historical CSV, labeled “at $x = 0$ ”. The source dictionary is deprecated.	Matches the rounded bar minimum, not the center entry below.
53.36 ps	Historical <code>RessigmaxzER0ejB</code> , generated from the archived optimal-estimator table.	Historical center entry; corrected-physics validation pending.
52.07 ps	Historical <code>RessigmaMINejB</code> , same source family.	Historical bar minimum; corrected-physics validation pending.

These are quoted source objects, not timing claims for the corrected detector.

v6 source snapshots at 3dc2c37: `sources/historical_napkin.py`, `sources/historical_results_macros.tex`. Pre-existing CSV snapshots have recorded SHA-256, not invented Git/run attribution. Historical EJ materials, END-only context; original N/seeds/physics commit are not established by these literals. See `sources/claims_audit` and `sources/provenance.json`.

Timing resolution needs a complete corrected-physics campaign

The historical timing results cannot be transferred by rescaling the light yield. No corrected σ_t value exists in this presentation.

- Preserve the historical deck and its datasets as provenance records.
- Specify geometry, material, SiPM/PDE data and estimator before any future timing campaign.
- Archive the source commit, seeds, event count and run command with each result.
- Keep measurement, source reproducibility and physical interpretation distinct.

Current scope: the two measured transport fixes and the archived light-collection studies. `presentations/v6/` is unchanged; no historical number is replaced by this deck.

Commit	Verified implementation change
bd78211	Explicit air gap; select a metal air-wrap reflector on that lineage.
4cb2c23	Port the explicit-gap geometry to EndTop; retain the metal reflector.
f3a8062	Select the dielectric <code>CreateBarSkinReflector()</code> at the air-wrap border.
87795b4	Merge the branch carrying that selection into main.
c7acb7a	Change the dielectric factory's input coefficient to 0.98; surface type stays dielectric.
f4d90a6	Restore the metal factory at the air-wrap border with input 0.98.

Genealogy is not experimental validation

The explicit-gap commit did not introduce the wrong factory selection. The later factory switch did. Author comments and commit messages are checked against diffs.

Each SHA checked with `git show --stat`; source diffs and ancestry recorded in EXEC_32. Basis: docs/DEFECTS_AND_CORRECTIONS.md. No event sample applies to a Git ancestry claim.

Commit	Verified change and qualification
51ca4ff	Al-foil metal surface and explicit wrap/world borders.
af5ddb7	Dielectric polished bar-skin design intended to preserve TIR; no surface MPT.
9b8361f	Add a reflectivity property to that dielectric factory. The old non-TIR recovery interpretation is not established for the later gap geometry.
fc5d8b0 / 76b582c	Corresponding EndTop factory ports on different lineages, not two sequential main ancestors.

Earlier skin-only decisions and the later explicit air-wrap border are different model placements. A surface property cannot be interpreted independently of that placement.

Verified SHA stat/diffs; branch qualifications follow docs/DEFECTS_AND_CORRECTIONS.md. Historical implementation intent is not a newly measured physical result.

For dielectric/dielectric, ordinary polished finish in the tested build:

- A draw above REFLECTIVITY + TRANSMITTANCE selects absorption.
- The intermediate interval selects direct transmission.
- The remaining ordinary-polished branch calls the dielectric/Fresnel calculation.

Thus the input coefficient is not directly an observed reflected fraction. The experimental check is the forward census, with nominal input 0.98 but observed reflection 13.371145% in R1. This source reading is labeled *hypothesis*; the measurement establishes the discrepancy. Repeated encounters are correlated, so no independent-Bernoulli significance is assigned.

Geant4 11.4.0: installed G4OpBoundaryProcess.cc source snapshot/hash under sources/. Experimental source: R1, 0336ba9, EJ-204/OPSC-101, EndTop $N_{TOP} = 70$, $x = 0$ mm, $N = 500$, seeds 26092601 8349041; sources/directional.

All rows: EJ-204/OPSC-101; vertical μ^- , 1 GeV; seeds 26092601 8349041. $x = 0$ mm.

Run	Mode / N_{TOP}	N	Workers	Source commit
R1	EndTop / 70	500	1	0336ba9
R2	EndTop / 70	500	1	218241a
A0	EndTop / 70	2000	1	a7a9090
B0	EndTop / 70	2000	1	f4d90a6
D0	END-only / 0	2000	1	2ddf205
D1	EndTop / 70	2000	1	2ddf205
D2	EndTop / 70	2000	1	8b64721
V1	END-only / 0	2000	12	2e3872e
D3	END-only / 0	2000	12	391fa4e

The multiworker rows use eventModulo 1; EXEC_30 G0 validated V1 against the one-worker END-only control. Source metadata, complete SHAs, commands, CWD, macro/binary hashes and sidecar copies: `sources/provenance.json`. No generic MT reproducibility claim is made.

END light-yield convention

Per event, $N_{\text{pe/end}} = \frac{1}{2}(N_L + N_R)$. Means and SEMs include zero-hit events. Ratio SEMs use the recorded zero-covariance approximation.

- Forward reflection fractions count eligible air $\rightarrow$ wrap encounters; excluded bookkeeping statuses remain in the source tables.
- Multiplicities use all-optical encounters divided by started optical tracks. They are not track-by-track bounce histories.
- Position scan: EndTop $N_{\text{TOP}} = 70$, $N = 1000/\text{arm/position}$; $x = \pm 200, \pm 500, \pm 650$ mm; seeds 26092601 8349041; one worker.
- Each figure has CSV, ROOT and metadata sidecars. The numeric macro file and the value register share a generator.

results_macros_v7.tex; sources/value_provenance.csv; sources/provenance.json; scripts/build_evidence.py. Per-cell source commits for the position arms: a7a9090/f4d90a6.